DO NOW — RECALL

Lesson 140 · Referential Integrity, Indexing and Keys

Name: _____ Class: _____ Date: _____

1. What is the result of UPDATE without a WHERE clause?

2. What is a trace table used for?

3. How many bits are in one byte?

4. What data structure uses LIFO?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Every row in the target table can be updated.

2. Model answer

Tracking variable state changes as an algorithm runs.

3. Model answer

Eight.

4. Model answer

Stack.

By the end, you can…

• Order.CustomerID references Customer.CustomerID.

• Invalid FK values are rejected unless permitted rules apply.

• An index can speed lookup at the cost of extra storage/write overhead.

Vocabulary — know these terms

referential integrity — Foreign-key values correspond to existing referenced keys or permitted NULLs.

index — Auxiliary structure helping the DBMS find rows efficiently.

primary key — Unique identifier of each record.

foreign key — Attribute referencing another relation.

NEW LEARNING

Read, annotate and explain

Link the records

Order.CustomerID references Customer.CustomerID.

Enforce integrity

Invalid FK values are rejected unless permitted rules apply.

Improve searches

An index can speed lookup at the cost of extra storage/write overhead.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integrity.

Customers 7 and 9

Identify the relevant information and make a prediction.

Order requests ID 12

Explain the change using correct technical terms.

Reject orphan reference

The DBMS rejects the foreign key value 12, unless a matching Customer row is created or the reference is changed. Policies for deleting parent rows may RESTRICT, CASCADE or SET NULL, where permitted and configured.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain referential integrity using Borrower and Loan with a suitable example. [6]

Question 2

Identify two benefits and one cost of indexing a frequently searched field. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A library deletes Borrower ID 8 while Loan rows refer to 8. Compare RESTRICT and CASCADE and justify a safe policy. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An index guarantees every column value it indexes is unique.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.